

Faculdade de Engenharia e Faculdade de Ciências da
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FDS/IDS Assignment

Predicting Cardiac Diseases in Children

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Introduction

Cardiovascular diseases in children require early diagnosis for effective intervention. In many developing regions, however, access to specialized pediatric cardiology is limited, forcing general practitioners to rely on basic history and physical exams for triage. This creates a bottleneck where missed diagnoses can be fatal, while unnecessary referrals overwhelm the healthcare system.

The primary diagnostic tool for heart defects is cardiac auscultation (listening for murmurs), but its accuracy depends heavily on the clinician's skill. This project explores whether Data Science can identify the most predictive clinical variables for pathology. We also investigate a critical question from previous research (Ferreira et al., 2018): *Is it possible to predict cardiac disease using only demographic and history data, bypassing the need for a physical exam?*

Data Description

In terms of data utilised, this is the description:

- **Continuous Data**
 - 'Peso';
 - 'Altura';
 - 'IMC' (derived both previous features);
 - 'Idade' (calculated according to 'DN' and 'Atendimento', to account for both temporal extraction, and, quality and consistency);
 - 'PA Sistolica';
 - 'PA Diastolica';
 - 'PPA'; ('PA Sistolica' / 'PA Distolica')
 - 'FC' (cardiac frequency).
- **Binary Data**
 - 'Sexo';
 - 'NORMAL x ANORMAL'; -> **this is the label for each data point**
- **Categorical Data (One-Hot Encoded)**
 - 'B2';
 - 'HDA1';
 - 'HDA2';
 - 'MOTIVO1';
 - 'MOTIVO2'.

Data Treatment

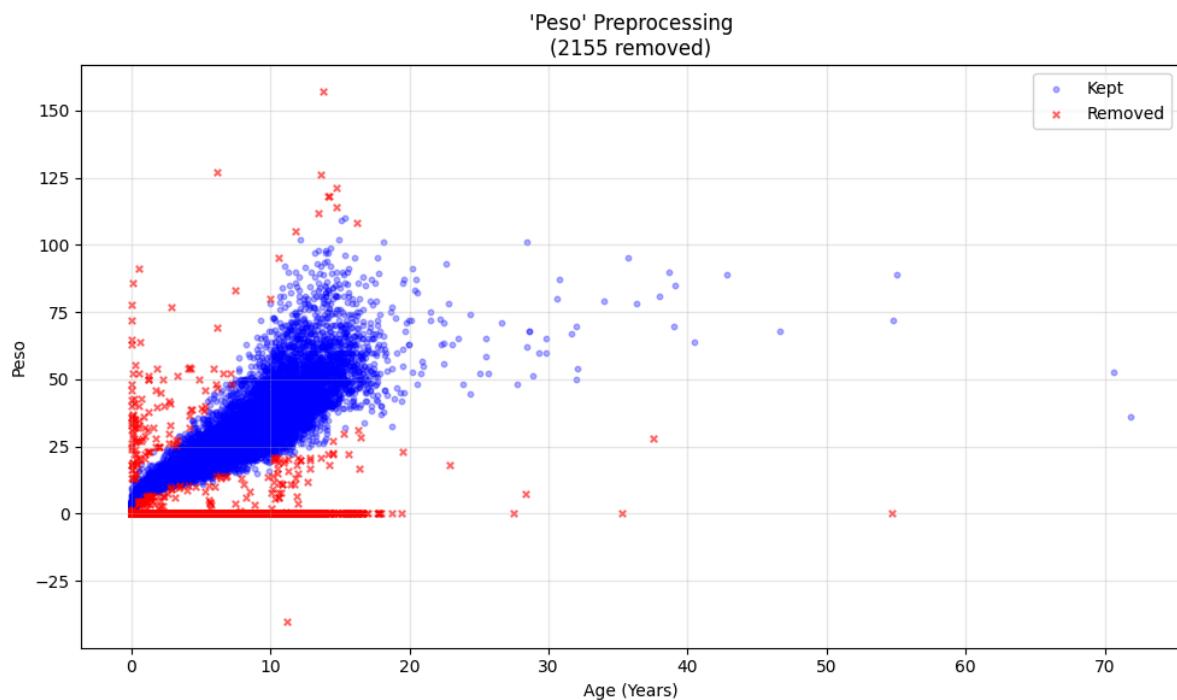
'Idade' negative and null value rows were removed. A calculated age value was obtained through the difference between date of 'Atendimento' and of 'DN' (date of birth), for values smaller than 1 year, we trusted the provided value, for other values (including cases where 'Idade' was null) we replaced the age value with the calculated one.

When 'Idade' was guaranteed, we grouped our data by age (by rounding to the nearest integer on each row), in order to remove 'Peso' and 'Altura' outliers.

'Peso' negative, null, and above 250kg values were instantly clipped, we also defined a minimum viable weight for each age group, to remove the lower bound outliers utilising real world data (W.H.O. 3 S.D. was the minimum chosen per age group).

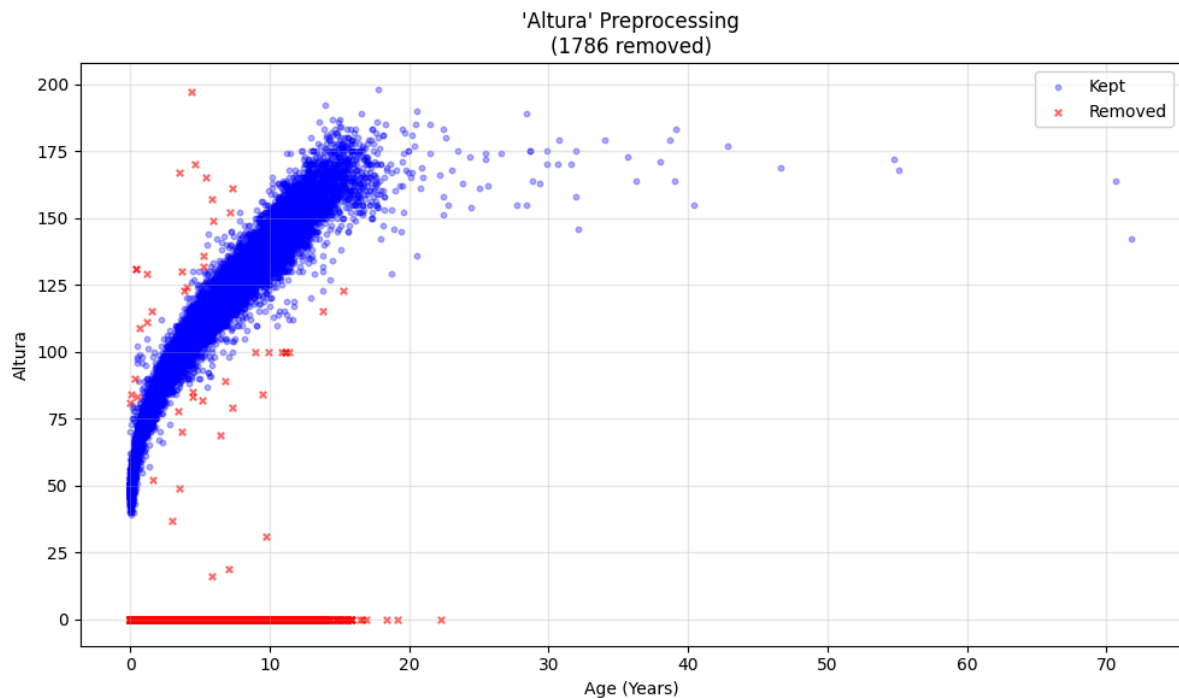
```
if age < 1: return 1.5 # Premature/Newborn
if age < 3: return 7.0 # Toddler
if age < 6: return 10.0 # Young Child
if age < 10: return 15.0 # School age
if age < 14: return 22.0 # Pre-teen
return 30.0 # Teen/Adult
```

On top of this, IQR based outlier-removal per age group was done, the multiplier chosen for the lower bound was 1.0 to account for our previous minimum viable weight, and for children with chronic conditions that might be very underweight, and upper bound 3.0, since we want to prioritise not eliminating obese and morbidly obese children, who are more prone to cardiac diseases and have higher representation and risk.

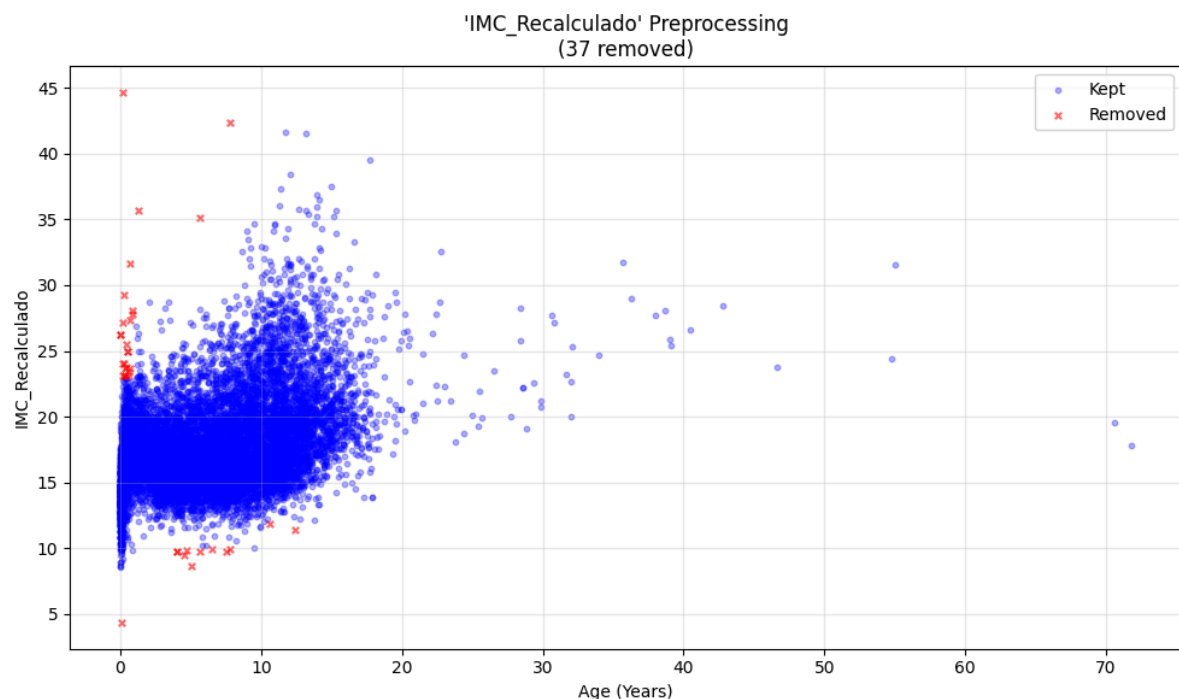


'Altura' followed the same logic, we removed values outside of the 15-250cm range instantly. We also utilised the IQR-based removal, with 2.0 being the chosen multiplier, to

also account for higher variability of height among children and patients with different afflictions.

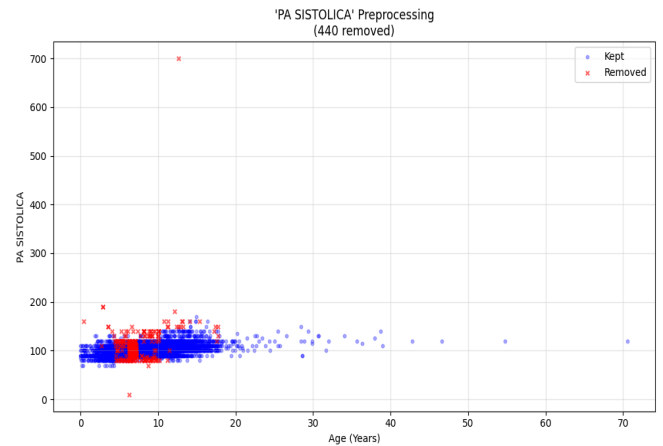
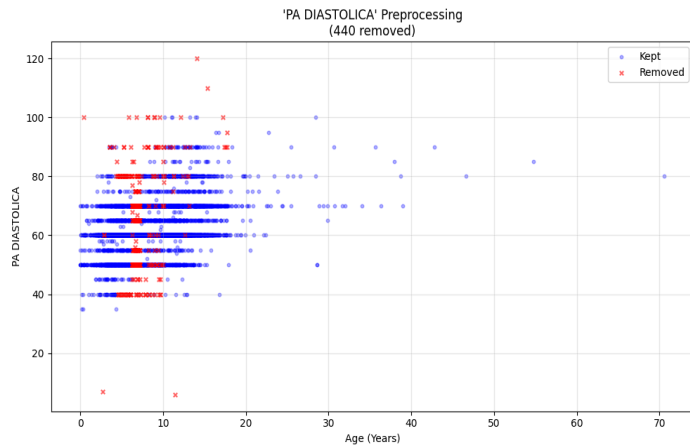


'IMC' was then recalculated and suffered new scrutiny. Infants (<1 year) had to have 'IMC' between 6 and 23, Toddlers/Kids (1-10 years old), between 10 and 35, and Teens/Adults (11 onwards) between 12 and 50.

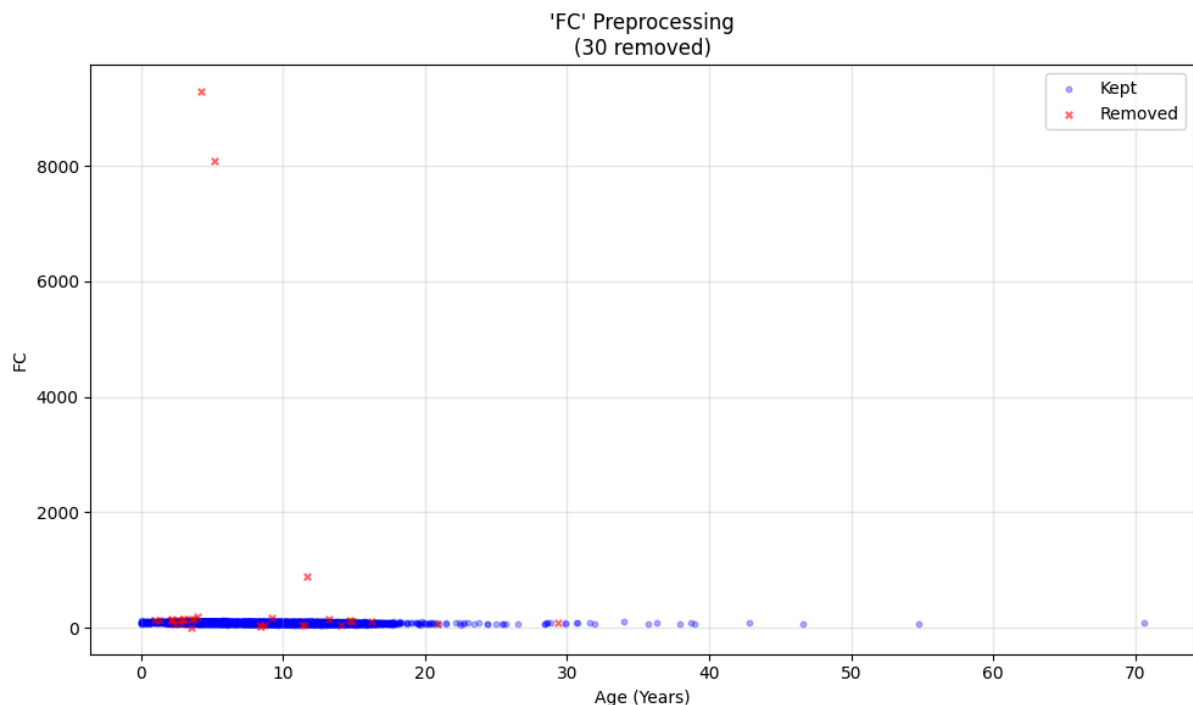


'Sexo' was also standardised, and different denominations were all substituted with a simple binary label (0 for Female and 1 for Male, this is essential for Machine Learning applications).

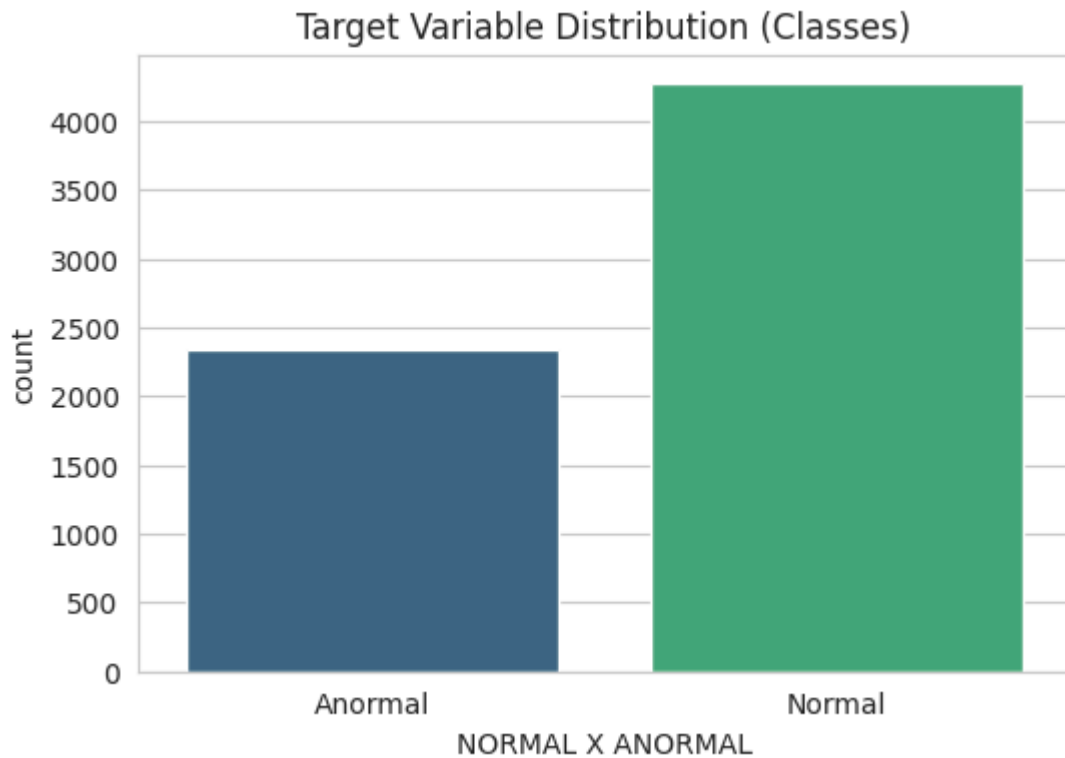
'PA SISTOLICA' and **'PA DIASTOLICA'** (treated at the same time due to their close relationship) were enforced as larger than 0, with diastolic having to be larger than systolic, and these two values not being permitted to have less than 10 units of distance between each other, all for medical reasons. Systolic also had to be between 40 and 320, and Diastolic between 20 and 200. Both utilised the same IQR-based outlier removal by age group, with multipliers of 2.0 and 3.5 for lower and upper bounds.



'FC' processing yielded a very low count of removals, even though we applied the same IQR-based outlier strategy with 2.0 and 3.5 as lower/upper bound multipliers, respectively, and enforced values between 25 and 300, medical limits. This suggests we are processing the data correctly, since **'FC'** ended up being auto-adjusted to our criteria after the previous steps.



Basic Description



The analysis of the target variable (**NORMAL X ANORMAL**) reveals a class imbalance. The dataset consists of approximately **64.6% Normal** cases and **35.4% Abnormal** cases.

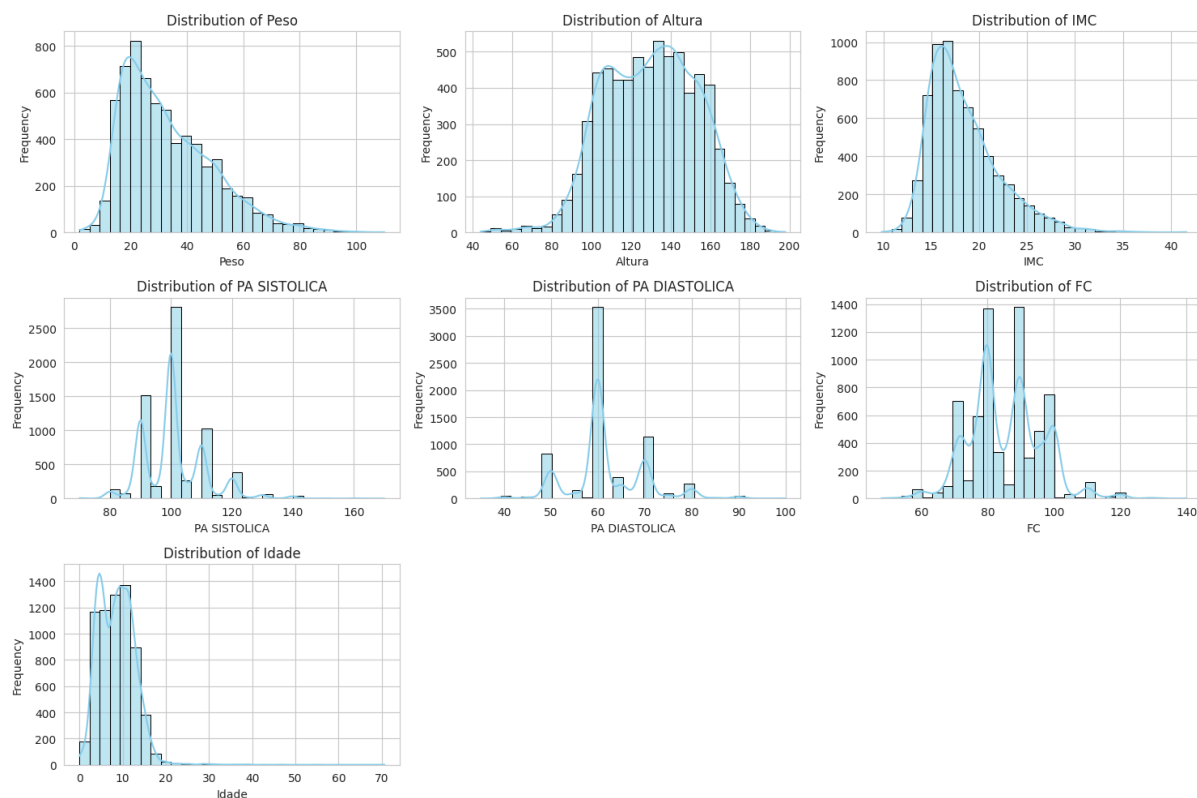
While the dataset is not severely imbalanced, the 2:1 ratio suggests that accuracy alone may be a misleading metric (a model could predict 'Normal' for everyone and still get 65% accuracy).

	count	mean	std	min	25%	50%	75%	max	IQR
Peso	6621.0	33.0	15.9	1.9	20.0	29.8	43.0	110.0	23.0
Altura	6621.0	130.0	23.1	44.0	111.0	131.0	148.0	198.0	37.0
IMC	6621.0	18.4	3.7	9.8	15.7	17.6	20.2	41.6	4.5
PA Sistólica	6621.0	100.8	10.2	70.0	90.0	100.0	105.0	170.0	15.0
Pa Diastólica	6621.0	62.0	8.0	35.0	60.0	60.0	65.0	100.0	5.0
FC	6621.0	85.5	10.9	48.0	78.0	84.0	92.0	140.0	14.0
Idade	6621.0	8.6	4.2	0.0	5.2	8.6	11.4	70.6	6.2

The statistical summary (Table 1) highlights the demographic spread of the pediatric patients.

The standard deviations for Weight (**Peso**) and Height (**Altura**) are large, reflecting a wide age range from infants to adolescents.

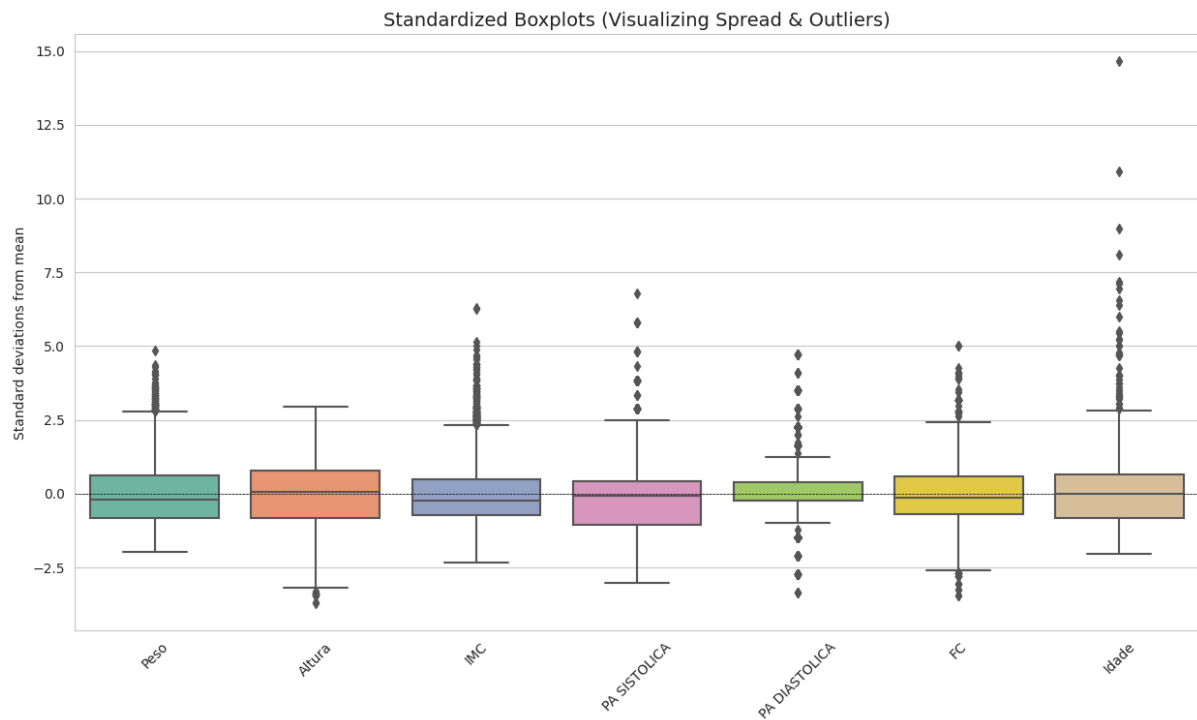
The Heart Rate (**FC**) and Blood Pressure (**PA**) also show significant variance, which is expected as these vitals change drastically as a child grows (e.g., infants have higher heart rates than teenagers)."



The histograms reveal that the demographic variables (Age, Weight, BMI) do not follow a normal (Gaussian) distribution; they are right-skewed.

Linear models (like Logistic Regression) often assume normal distributions and might perform poorly on this raw data. However, **Decision Trees** are robust to non-normal data and do not require transformations like log-scaling.

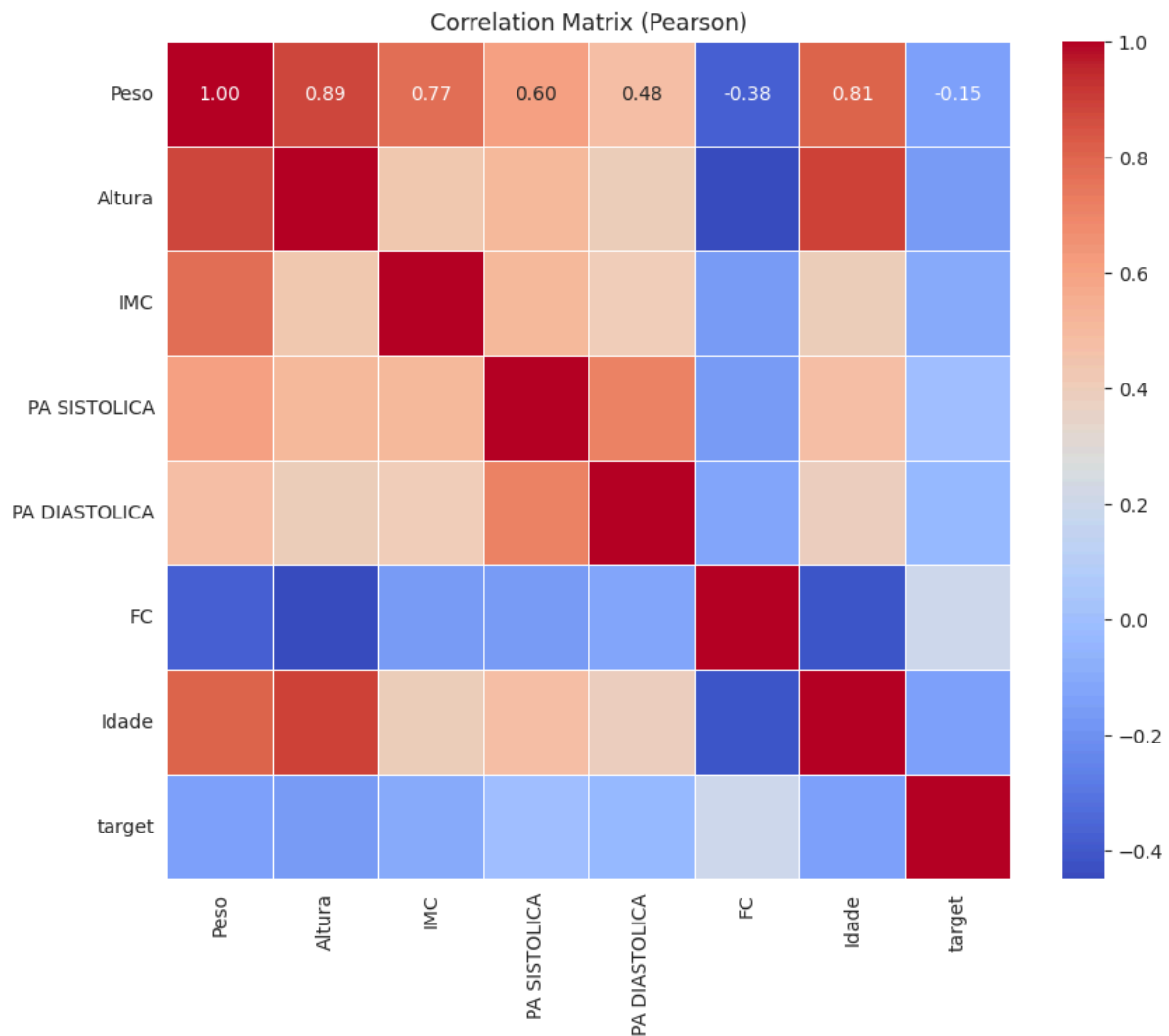
The wide spread in Heart Rate (**FC**) and Blood Pressure confirms that the dataset contains distinct physiological groups (infants vs. adolescents), suggesting that age is a confounding factor that the model must learn to separate.



The standardized boxplots identify significant statistical outliers, particularly in **Systolic Blood Pressure (PA SISTOLICA)** and **BMI (IMC)**

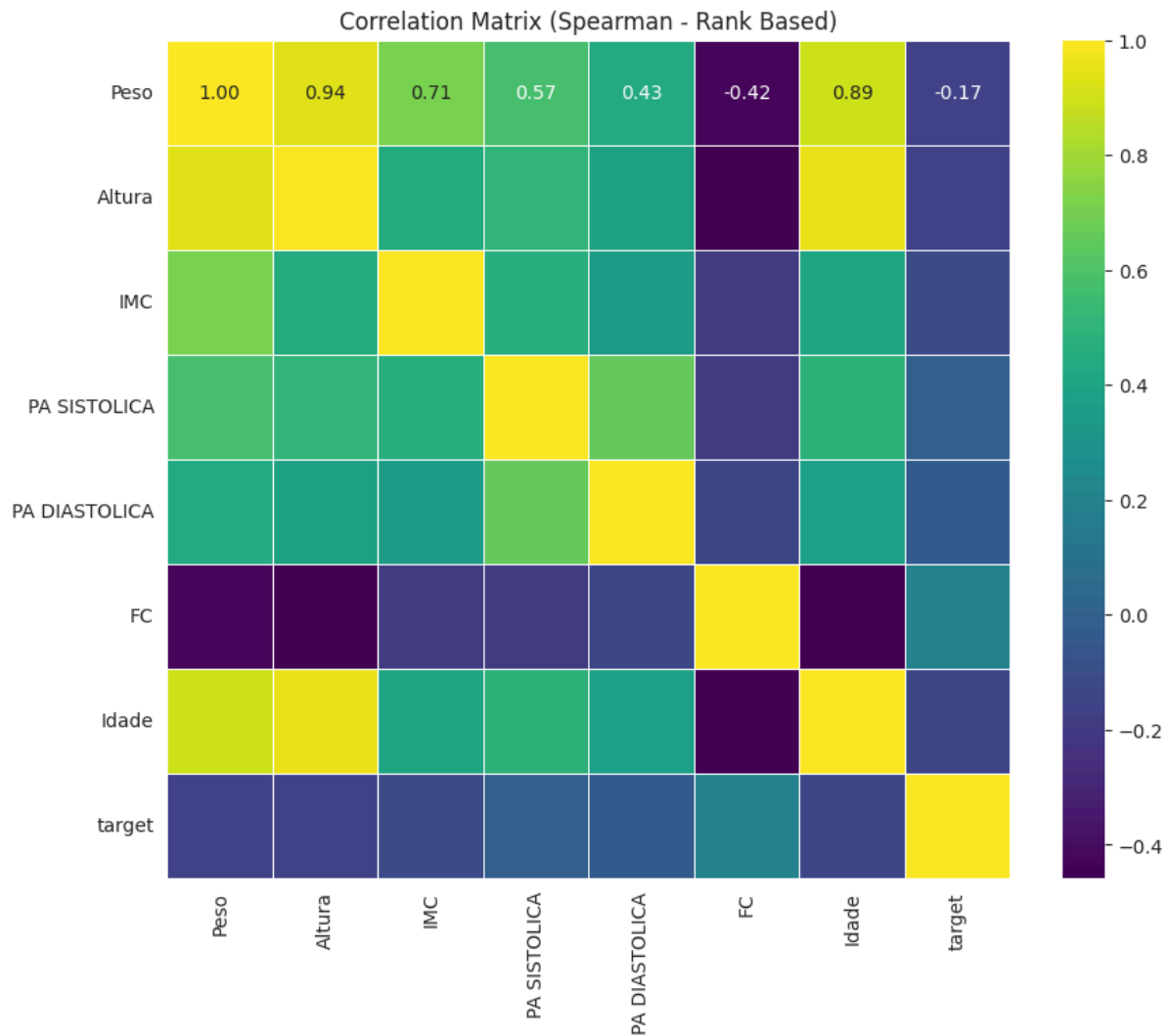
- In many datasets, outliers represent measurement errors to be removed. However, in this cardiac dataset, high blood pressure and high BMI are known risk factors for heart pathology. These 'outliers' represent the most critical patients, those with hypertension or obesity.
- We decided **not to remove** these outliers. Removing extreme values in **PA SISTOLICA** would effectively delete the hypertensive patients we aim to diagnose. Instead, we rely on the robustness of the Random Forest/Decision Tree algorithms to handle these extremes.

Bivariate Analysis



The correlation analysis highlights strong multicollinearity among anthropometric variables, specifically between Height and Weight ($r=0.89$) and Height and Age ($r=0.89$). This redundancy is expected as children naturally grow in both dimensions over time¹.

However, BMI (**IMC**) shows a much weaker correlation with Age ($r=0.40$), indicating that it provides distinct information regarding nutritional status rather than just developmental growth. Therefore, to avoid redundancy in the predictive models, we will discard Weight and Height and retain BMI and Age as independent predictors.



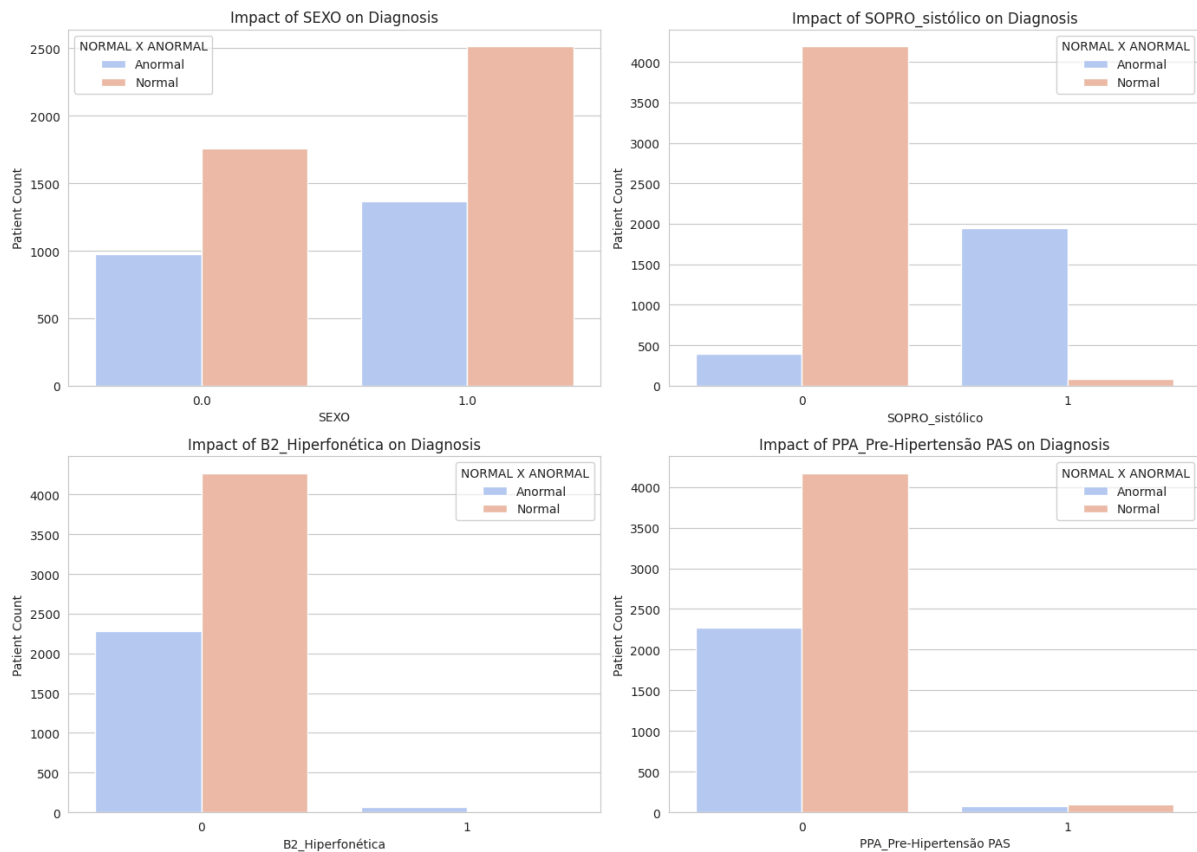
To ensure the robustness of our correlation analysis, we complemented the Pearson method (which measures linear dependence) with the Spearman Rank Correlation. This was crucial because biological variables, such as growth metrics, often exhibit non-linear but monotonic relationships that Pearson might underestimate.

However, the Spearman coefficients yielded values nearly identical to the Pearson matrix.

The correlation between Height and Weight remained extremely high (~ 0.89), and the relationship between Age and Height was equally strong.

The similarity between these two metrics confirms that the relationships among these anthropometric variables are not only strong but also highly linear within this specific pediatric population.

This robust redundancy reinforces our feature selection strategy. Since both linear and rank-based methods detect the same overlap, we can confidently exclude Weight and Height to prevent multicollinearity, relying instead on BMI (which showed a distinct, lower correlation profile) and Age as our primary independent predictors.



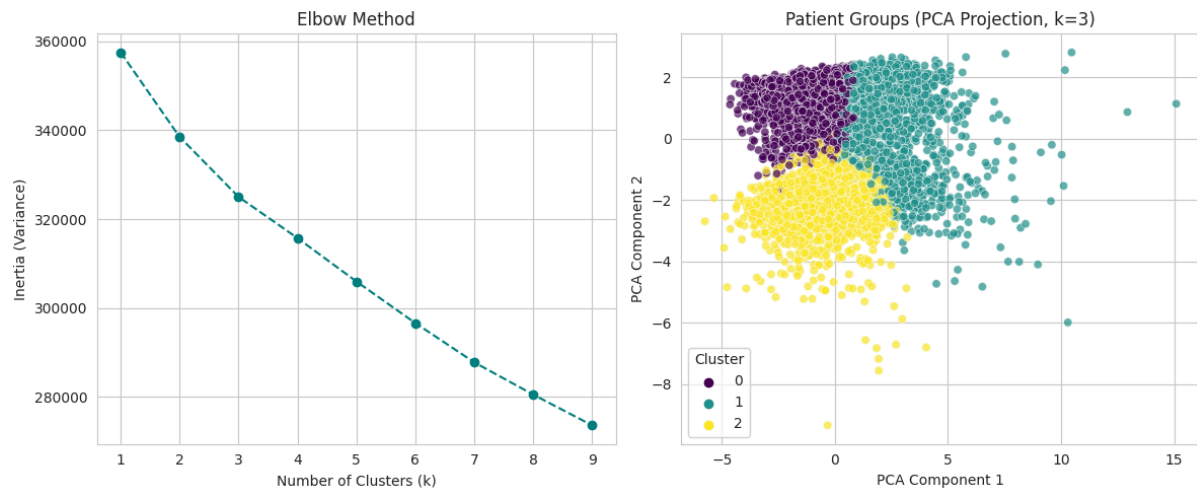
This visualization confirms that **Systolic Murmur (SOPRO_sistólico)** is the single most dominant indicator of cardiac pathology in our dataset.

- The chart reveals a stark contrast where the vast majority of patients without a murmur are classified as 'Normal', whereas the presence of a murmur is strongly associated with the 'Abnormal' (Pathology) class.

The analysis of the **Second Heart Sound (B2)** suggests it is a specific risk factor, but the findings must be interpreted with caution.

- While 'Hyperphonic B2' is strongly associated with the 'Abnormal' group, the absolute number of patients presenting this symptom is very low compared to the total population.
- A hyperphonic B2 is typically a sign of severe hemodynamic changes like Pulmonary Hypertension, making it a highly specific marker for pathology. However, due to the small sample size of patients with this specific finding, our ability to generalize its predictive power is limited. It serves as a rare but serious indicator rather than a common screening tool.

Multivariate Analysis

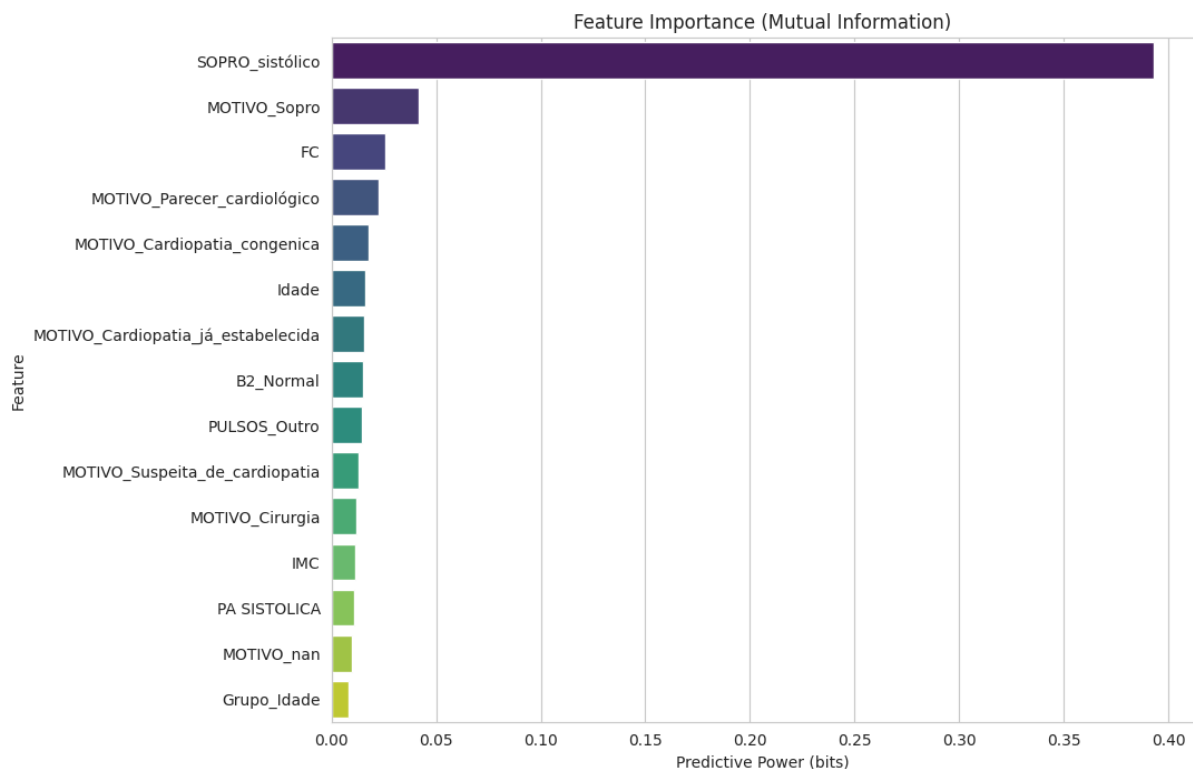


	Idade	IMC	PA SISTOLICA	PA DIASTOLICA	target	SOPRO_sistólico	PPA_Pre-Hipertensão PAS
Cluster							
0	6.339839	17.013452	96.184721	58.665724	0.282767	0.227971	0.016572
1	12.743717	21.203116	108.602081	67.303271	0.229011	0.179881	0.036174
2	7.332686	17.335864	98.723344	60.643964	0.555712	0.517614	0.026303

The K-Means analysis (k=3) successfully segmented the patients into three distinct profiles based on developmental stage and clinical risk:

- **Cluster 0 (Young & Low Risk):** Represents young children (Mean Age ~6.3) with normal physiological parameters and a low rate of pathology (28%).
- **Cluster 1 (Healthy Adolescents):** Represents the oldest group (Mean Age ~12.7). Although they have higher Blood Pressure (108/67) and BMI, this is physiological for their age. They have the lowest risk of pathology (22.9%) and the fewest murmurs.
- **Cluster 2 (Symptomatic / High Risk):** Represents the critical group for diagnosis. While similar in age to Cluster 0 (~7.3 years), this group has a drastically higher pathology rate (55.6%). This cluster is defined by clinical symptoms, specifically the high prevalence of Systolic Murmurs (51.8%), compared to <23% in the other groups.

The clustering confirms that the dataset contains latent structures where patients are stratified primarily by **Age** (Growth) and **Murmurs** (Pathology).



The Mutual Information ranking reveals which variables share the most information with the target (Pathology), capturing both linear and non linear relationships.

The **Systolic Murmur (SOPRO_sistólico)** is the overwhelming driver of diagnosis, with a score of **0.39**. This dwarfs all other features, confirming that the physical detection of a murmur is the primary signal for cardiac pathology in this dataset.

Several top features relate to the **Reason for Consultation (MOTIVO)**, such as *Referral for Murmur* or *Suspected Heart Disease*. This indicates that the initial medical triage is highly effective. Patients referred for specific cardiac concerns are statistically much more likely to be sick.

Interestingly, **Heart Rate (FC)** ranks higher (0.026) than **Age (Idade)** or BMI. This suggests that hemodynamic stability (vital signs) is a more direct indicator of pathology than simple growth metrics like age or weight.

Predictive Models

	Accuracy	Sensitivity (Recall)	Specificity	F1-Score	ROC-AUC
Model					
Decision Tree	0.926411	0.833570	0.977361	0.889226	0.915504
SVM (RBF)	0.921371	0.829303	0.971897	0.881997	0.927761
Naive Bayes	0.693548	0.159317	0.986729	0.269231	0.910697
Random Forest	0.927923	0.837838	0.977361	0.891749	0.943150

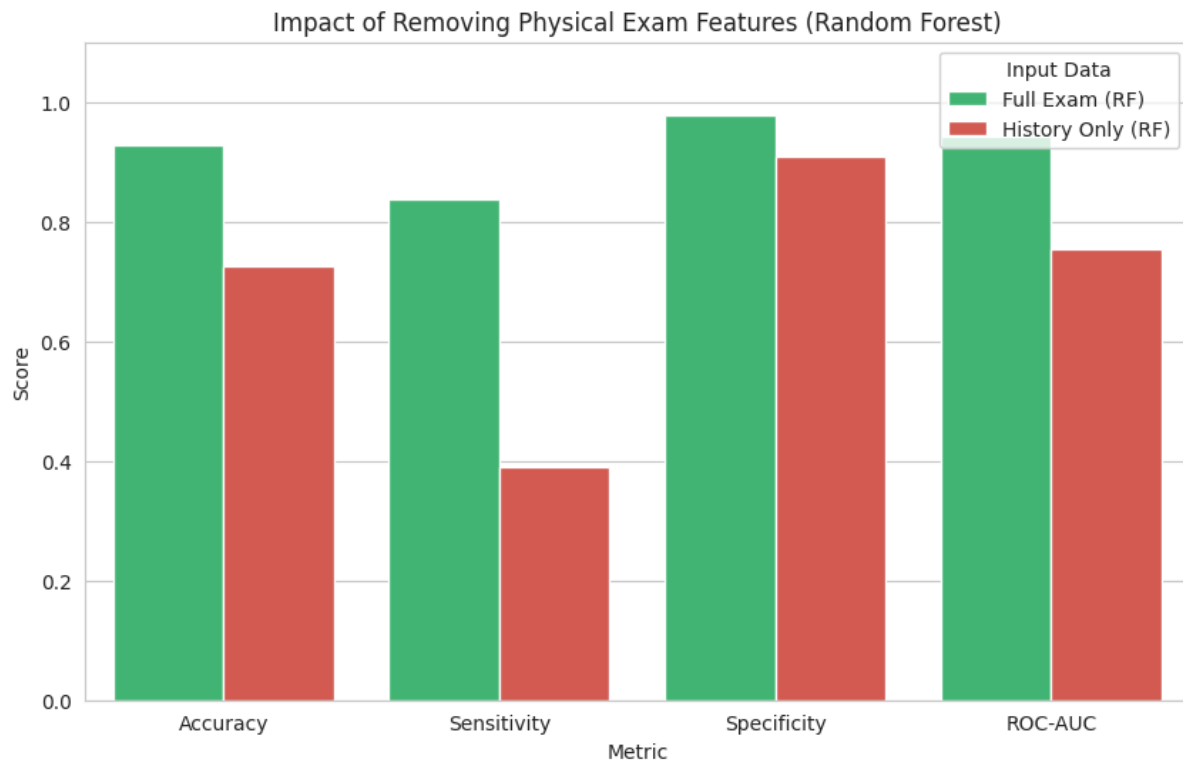
The supervised learning models achieved remarkable performance, with the top classifiers reaching accuracies above 92% and ROC-AUC scores exceeding 0.94. This high level of success is largely driven by the inclusion of the **Systolic Murmur (SOPRO)** variable, which our feature selection phase identified as the dominant predictor of pathology.

Model Comparison:

- **Random Forest (The Statistical Winner):** The Random Forest model achieved the highest overall metrics, with an **Accuracy of 92.8%** and an **ROC-AUC of 0.94**. It also demonstrated the highest **Sensitivity (83.8%)**, meaning it was the most effective at correctly identifying sick children. Its ability to aggregate multiple decision trees likely helped it capture subtle interactions between the murmur and secondary signs like heart rate or BMI.
- **Decision Tree (The Explainable Alternative):** The single Decision Tree performed almost as well as the Random Forest (Accuracy 92.6%), with only a negligible drop in Sensitivity. While the Random Forest is statistically superior, the Decision Tree offers the advantage of **interpretability**. Medical practitioners can trace the specific rules (e.g., "If Murmur Present, Check Blood Pressure") used to reach a diagnosis, which builds trust in the clinical setting.
- **Naive Bayes (The Outlier):** The Naive Bayes model showed a critical failure in **Sensitivity (15.9%)**. Although it had extremely high Specificity (98.7%), it essentially classified nearly everyone as "Healthy." This suggests that the assumption of feature independence is too strong for this dataset. The model failed to leverage the complex dependencies between clinical history and physical findings that the tree based models successfully exploited.

With the physical examination data included, both the Random Forest and Decision Tree are highly effective **Diagnostic Assistants**. They can reliably confirm a diagnosis when a doctor inputs clinical findings like a murmur. However, their reliance on this single powerful feature suggests they are automating the final step of the diagnosis rather than predicting risk from history alone.

	Accuracy	Sensitivity (Recall)	Specificity	F1-Score	ROC-AUC
Model					
DT (No Murmur)	0.715222	0.345661	0.918033	0.462417	0.732407
SVM (No Murmur)	0.724798	0.394026	0.906323	0.503636	0.720147
RF (No Murmur)	0.724798	0.389758	0.908665	0.500914	0.752992



To evaluate the feasibility of a "History Only" screening tool (e.g., for use in telemedicine or by non-specialists without a stethoscope), we conducted an ablation experiment. We trained the same predictive models after removing all variables related to the physical heart examination (Murmurs).

Results: The removal of the murmur ([SOPRO](#)) variables caused a catastrophic drop in model performance across all algorithms.

- The most alarming result was the drop in Sensitivity (Recall). While the full models achieved ~83% sensitivity, the "History Only" models plummeted to **below 40%** (34.6% for Decision Tree, 39.0% for Random Forest).
- A sensitivity of 39% means that a screening tool based solely on demographics and clinical history would miss approximately **6 out of every 10 sick children**. This is an unacceptable failure rate for a medical diagnostic system.
- Even the **Random Forest**, which is powerful at finding complex non linear patterns, failed to recover the lost performance (Sensitivity 39.0%, ROC-AUC 0.75). This confirms that the predictive signal for cardiac pathology is not "hidden" in variables like Age, BMI, or Heart Rate. It is explicitly tied to the physical findings.

We conclude that demographic factors and clinical history alone are **insufficient** for accurate diagnosis. The physical auscultation (listening to the heart) remains the critical diagnostic step. Therefore, any AI solution deployed in this domain must be designed as an **Assistant to the Physical Exam** (Triage Support) rather than a standalone replacement for it.

Conclusion

This project set out to identify the most predictive variables for pediatric cardiac pathology and to assess the feasibility of automated screening. Our analysis, performed on a dataset of clinical, yielded three critical insights:

1. **Data Quality:** Rigorous preprocessing, including the removal of physiological outliers and the standardization of categorical labels, was essential for model stability. We found that deriving **BMI (IMC)** and discarding raw Height/Weight eliminated significant multicollinearity ($r=0.89$) while preserving nutritional information.
2. **The Dominant Predictor:** Consistent with previous studies (Ferreira et al., 2018), our Feature Importance analysis (Mutual Information) identified **Systolic Murmur (SOPRO)** as the single most informative variable. Its presence is the primary "red flag" used for diagnosis, dwarfing the predictive power of demographic factors like Age or Sex.
3. **The Limits of Automation:** Our ablation study directly addressed the question of "history-only" screening. We demonstrated that removing physical exam findings causes a catastrophic drop in model **Sensitivity** (from ~83% to <40%). This proves that demographic data alone carries insufficient signal for reliable diagnosis.

Final Recommendation:

Artificial Intelligence cannot currently replace the physical examination in this domain. The high performance of our Random Forest model (ROC-AUC 0.94) relies on the doctor's ability to detect a murmur first. Therefore, we recommend deploying these models as Clinical Decision Support Systems (CDSS) to verify diagnoses and prioritize high-risk referrals, rather than as standalone screening tools for non-specialists.

References

- [1] P. Ferreira, T. T. V. Vinhoza, A. Castro, F. Mourato, T. Tavares, S. Mattos, I. Dutra, and M. Coimbra, "Knowledge on Heart Condition of Children based on Demographic and Physiological Features," *2018 IEEE 31st International Symposium on Computer-Based Medical Systems (CBMS)*, Karlstad, Sweden, 2018.
- [2] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2011.
- [3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.
- [4] World Health Organization (WHO), "Child Growth Standards," [Online]. Available: <https://www.who.int/tools/child-growth-standards>.